

Definition 3.3: Orthogonal Projection

Write $\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$, where $\vec{u}_{\parallel} \parallel \vec{v}$ and $\vec{u}_{\perp} \perp \vec{u}_{\parallel}$. Then, $\vec{u}_{\parallel}$ is called the **orthogonal projection** of $\vec{u}$ onto $\vec{v}$.